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CSE-302: Course Project Part 2

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Dataset title: Overwatch hero statistics

Source: [Kegggle](#).

The variables I chose are: Hero, Win_rate, Rank

This is the same dataset I used in part 1. I chose this specific dataset because it's relevant to my favorite video game, "Overwatch 2". This is a game I'm deeply knowledgeable on, and love growing deeper intuition on it. Win rates with specific playable characters or "heroes" are really important to know which characters perform well at a point in time, and which don't. It basically tells you which heroes are better to win with. Another factor is a player's individual rank, since some heroes could be good in one rank, but worse in the other.

Application 1: Binary Search Tree

The Overbuff records could be stored in a Binary Search Tree sorted by the win rate percentage. This would allow the program to organize hero records from lowest to highest win rate. If a user wanted to know which heroes in ranks had a win percentage above a certain number, the tree could be searched and printed using reverse-order traversal. For people that play Overwatch, this information would be useful for identifying "meta heroes", meaning heroes that have the highest

performance. It would also help users analyze which heroes perform in a certain rank compared to others.

Application 2: Hash Map

A hash table could be used to map each hero name to all the records belonging to that hero. For example, the key could be the hero name and accessing that would show only that heroes win rate at different ranks. This would make it fast to look up a specific hero without scanning the entire dataset. A player could enter a hero name and quickly see how that hero performs across rank categories.

For my project I decided to use Application 1, using a BST to store my data and sort winrates greater than the user's input.